MF 281: Probability Theory

Week 6: The standard discrete distributions

Choosing one by argument, not by habit

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Lecture notes for students of MF 281 Probability Theory.

Week 6: The standard discrete distributions

Learning Objectives

By the end of this week you should be able to:

- state the mechanism behind the Bernoulli, binomial, geometric, negative binomial, Poisson and hypergeometric distributions;
- write down each mass function and use it;
- choose a distribution from a description of how the data arose;
- say when the binomial is wrong because sampling is without replacement;
- use the Poisson approximation to the binomial, and say when it is good.

6.1 Choose by mechanism

A distribution is not a formula you try until one fits. It is a claim about how the numbers were generated. If the mechanism matches, the distribution follows; if it does not, no amount of fitting rescues it.

If the question is	the distribution is
One trial, success or failure	Bernoulli
How many successes in n fixed independent trials	Binomial
How many trials until the first success	Geometric
How many trials until the rth success	Negative binomial
How many events in a fixed interval, at a constant average rate	Poisson
How many successes in n draws *without* replacement	Hypergeometric

The questions to ask of any dataset: is the number of trials fixed, or is the count of successes fixed? Is sampling with replacement? Are the trials independent, with a constant probability?

6.2 Bernoulli

One trial, $X \in \{0, 1\}$, with $P(X = 1) = p$. Its mass function is $p_X(1) = p$, $p_X(0) = 1 - p$, and $\mathrm{E}(X) = p$, $\mathrm{Var}(X) = p(1 - p)$, proved next week. Everything else in this chapter is built from it.

6.3 Binomial

n independent Bernoulli trials, each with the same p, and X the number of successes:

$$p_X(k) = \binom{n}{k} p^k (1 - p)^{n-k}, \qquad k = 0, 1, \ldots, n.$$

The factor $p^k(1 - p)^{n-k}$ is the probability of one particular sequence with k successes, and $\binom{n}{k}$ counts the sequences. Mean np, variance $np(1 - p)$.

Example 1: ten castings

Thirty percent of castings need rework, independently. Ten are produced.

Solution

$$P(X=4)=\binom{10}{4}(0.3)^4(0.7)^6=210(0.0081)(0.1176)=0.2001.$$

$$P(X\le 2)=(0.7)^{10}+10(0.3)(0.7)^9+45(0.3)^2(0.7)^8=0.3828.$$

On average $np=3$ need rework, with variance $np(1-p)=2.1$.

The four conditions

Fixed n; two outcomes per trial; constant p; independence. Drop any one and the binomial is wrong. The usual casualty is the last: castings from one mould after a tool wears are not independent, and the real distribution has fatter tails than the binomial predicts.

6.4 Geometric

Independent trials with probability p, counted until the first success:

$$p_X(k)=(1-p)^{k-1}p, \qquad k=1,2,3,\ldots$$

with $P(X>k)=(1-p)^k$ and mean $1/p$.

Example 2: the fifth theodolite

One theodolite in a hundred is defective. Items are inspected one at a time. What is the probability the fifth inspected is the first defective one?

Solution

$$P(X=5)=(0.99)^4(0.01)=0.0096.$$

On average you inspect $1/p=100$ before finding one, which is the sanity check: a one percent defect rate means about one in a hundred.

The memoryless property

For the geometric distribution,

$$P(X>m+k\mid X>m)=\frac{(1-p)^{m+k}}{(1-p)^m}=(1-p)^k=P(X>k).$$

Twenty sound items in a row tell you nothing about when the next defective one arrives. The geometric is the only discrete distribution with this property, and it is the gambler's fallacy formalised: the process has no memory even though people insist it does.

6.5 Negative binomial

Now wait for the rth success. With X the number of trials required,

$$p_X(k)=\binom{k-1}{r-1}p^r(1-p)^{k-r}, \qquad k=r,r+1,\ldots$$

The last trial must be a success, and the other $r-1$ successes fall anywhere among the first $k-1$ trials, which is where the binomial coefficient comes from. Mean r/p.

Example 3: drilling for two discoveries

An exploration company drills until it has made two discoveries. Each hole is independently successful with probability 0.1.

Solution

Exactly two holes:

$$p_X(2) = \binom{1}{1}(0.1)^2(0.9)^0 = 0.01.$$

Exactly three:

$$p_X(3) = \binom{2}{1}(0.1)^2(0.9)^1 = 2(0.01)(0.9) = 0.018.$$

Exactly five: $\binom{4}{1}(0.1)^2(0.9)^3 = 4(0.01)(0.729) = 0.0292$.
The expected number of holes is $r/p = 20$, which is the number the budget has to survive. Note that $p_X(3) > p_X(2)$: there are more ways to arrange the failure.

6.6 Poisson

Counts of events in a fixed interval of time or space, occurring at a constant average rate λ, independently of one another:

$$p_X(k) = \frac{e^{-\lambda}\lambda^k}{k!}, \qquad k = 0, 1, 2, \ldots$$

with mean and variance both equal to λ. It sums to one because $\sum_k \lambda^k/k! = e^{\lambda}$.

Example 4: breakdowns

A plant suffers breakdowns at an average of three per month.

Solution

$$P(X = 0) = e^{-3} = 0.0498, \qquad P(X = 2) = \frac{e^{-3}3^2}{2!} = 0.2240, \qquad P(X \le 4) = 0.8153.$$

So a month with no breakdown at all happens about once in twenty months, and four or fewer covers about 82% of months. Planning maintenance capacity for exactly three, the average, would be under-resourced in nearly half of all months.

6.6.1 Poisson as a limit of the binomial

Let $n \to \infty$ and $p \to 0$ with $np = \lambda$ fixed. Then

$$\binom{n}{k}p^k(1-p)^{n-k} = \underbrace{\frac{n(n-1)\cdots(n-k+1)}{n^k}}_{\to 1} \cdot \frac{\lambda^k}{k!} \cdot \underbrace{\left(1 - \frac{\lambda}{n}\right)^n}_{\to e^{-\lambda}} \cdot \underbrace{\left(1 - \frac{\lambda}{n}\right)^{-k}}_{\to 1} \longrightarrow \frac{e^{-\lambda}\lambda^k}{k!}.$$

Example 5: how good is the approximation

A batch of 1000 components each fails independently with probability 0.002. Compare the exact binomial with the Poisson approximation, $\lambda = np = 2$.

Solution

$$P(X = 3) \text{ exactly } = \binom{1000}{3}(0.002)^3(0.998)^{997} = 0.18063,$$

$$P(X = 3) \text{ approximately } = \frac{e^{-2}2^3}{3!} = 0.18045.$$

They agree to three decimal places. The rule of thumb is $n \ge 20$ with $p \le 0.05$, or $n \ge 100$ with $np \le 10$; here both hold comfortably. The approximation mattered more before computers, but

the limit also explains *why* rare events in large populations are Poisson: it is the natural law for many chances, each small.

6.7 Hypergeometric

Draw n items without replacement from N, of which K are defective. Then

$$p_X(k) = \frac{\binom{K}{k}\binom{N-K}{n-k}}{\binom{N}{n}}.$$

Here trials are *not* independent: removing a defective item changes what is left.

Example 6: a small batch, and why the binomial misleads

A box holds 20 items, 5 defective. Four are drawn without replacement.

Solution

$$P(X=1) = \frac{\binom{5}{1}\binom{15}{3}}{\binom{20}{4}} = \frac{5\times 455}{4845} = 0.4696, \qquad P(X=0) = \frac{\binom{15}{4}}{\binom{20}{4}} = 0.2817.$$

Treating it as binomial with $p = 5/20 = 0.25$ would give $\binom{4}{1}(0.25)(0.75)^3 = 0.4219$ for one defective: out by five percentage points.

When the binomial is safe

When the sample is a small fraction of the population, usually taken as $n/N \le 0.05$, the two agree closely, because removing a few items barely changes the composition. Here $n/N = 0.2$, far too large. This is the same finite-population correction that appears in sampling theory.

6.8 Putting it together

Distribution	Support	Mean	Variance	Fixed quantity
Bernoulli(p)	$0, 1$	p	$p(1-p)$	one trial
Binomial(n, p)	$0..n$	np	$np(1-p)$	number of trials
Geometric(p)	$1, 2, \ldots$	$1/p$	$(1-p)/p^2$	one success
Neg. binomial(r, p)	$r, r+1, \ldots$	r/p	$r(1-p)/p^2$	r successes
Poisson(λ)	$0, 1, \ldots$	λ	λ	the interval
Hypergeom.(N, K, n)	$0..n$	nK/N	see note	draws, no replacement

The hypergeometric variance is $n\frac{K}{N}\left(1-\frac{K}{N}\right)\frac{N-n}{N-1}$: the binomial variance times the finite population correction, the same factor you met in sampling.

Binomial and negative binomial are the same experiment asked two ways. Fix the number of trials and count successes: binomial. Fix the number of successes and count trials: negative binomial. Which one applies depends on what the experimenter decided in advance, and that is a fact about the protocol, not about the data.

6.9 Three ways to choose wrongly

1. **Binomial for sampling without replacement** from a small population. Use the hypergeometric, or check $n/N \le 0.05$ first.
2. **Poisson for a rate that is not constant.** Breakdowns cluster after a maintenance change, and arrivals peak at shift handover. A Poisson fitted across a varying rate understates the variance.

3. **Assuming independence because the formula needs it.** The need is not evidence. Say what makes it plausible, or use a model that does not require it.

6.10 Summary

Week 6 in six lines

- Pick a distribution from the mechanism, then check the conditions.
- Binomial: fixed trials, count successes. Negative binomial: the reverse.
- The geometric is memoryless, and it is the only discrete one that is.
- Poisson is the law of many chances, each small: λ is both mean and variance.
- Without replacement means hypergeometric, unless the sample is tiny relative to the population.
- Every one of these assumes independence somewhere; say why it holds.

Exercises

1. Name the distribution and its parameters for each: the number of sixes in 20 rolls; the number of rolls until the first six; the number of faulty bulbs in a sample of 6 drawn from a box of 40 containing 8 faulty; the number of accidents at a junction in a year.

2. Fifteen percent of items need rework. In a sample of eight, find $P(X \geq 2)$.

3. For the same process, find the probability that the first item needing rework is the fourth inspected, and the expected number inspected before one is found.

4. Derive the mean of the geometric distribution, $\mathrm{E}(X) = 1/p$, using $\sum_{k\geq 1} k(1-p)^{k-1} = 1/p^2$.

5. Faults occur at four per week on average. Find the probability of none in a week, exactly five in a week, and at least one in a week.

6. For the same process, what is the probability of exactly eight faults in two weeks? State the property of the Poisson that you used.

7. A drilling programme needs three discoveries, each hole succeeding with probability 0.2. Find the probability that exactly ten holes are required, and the expected number.

8. A batch of 50 contains 6 defective items. Five are drawn without replacement. Find $P(\text{no defective})$ and $P(\text{at least two defective})$.

9. For Exercise 8, compute the binomial approximation and state the percentage error. Is the approximation acceptable here?

10. A call centre receives 1200 calls a day, each independently a complaint with probability 0.004. Use a suitable approximation to find the probability of exactly six complaints, and justify the choice.

11. Show that for the geometric distribution $P(X > m + k \mid X > m) = P(X > k)$, and explain in one sentence what this means for a machine operator.

12. An engineer fits a Poisson to monthly breakdowns and finds the sample variance is three times the mean. What does that suggest, and what would you check?

Answers to the odd-numbered exercises

1. Binomial$(20, 1/6)$; geometric$(1/6)$; hypergeometric with $N = 40$, $K = 8$, $n = 6$, since the draw is without replacement from a small box; Poisson, with λ the average number of accidents per year.

3. $P(X = 4) = (0.85)^3(0.15) = 0.0921$. The expected number inspected is $1/0.15 = 6.67$, so about seven items.

5. With $\lambda = 4$: $P(0) = e^{-4} = 0.0183$; $P(5) = e^{-4}4^5/5! = 0.1563$; $P(\geq 1) = 1 - e^{-4} = 0.9817$.

7. Negative binomial with $r = 3$, $p = 0.2$: $P(X = 10) = \binom{9}{2}(0.2)^3(0.8)^7 = 36(0.008)(0.2097) = 0.0604$. Expected number of holes $= 3/0.2 = 15$.

9. Binomial with $n = 5$, $p = 6/50 = 0.12$ gives $P(0) = (0.88)^5 = 0.5277$ against the exact hypergeometric 0.5126, an error of about 3%. Here $n/N = 0.1$, twice the usual threshold, so the approximation is tolerable for a rough figure and not for a contractual one.

11. $P(X > m + k \mid X > m) = (1 - p)^{m+k}/(1 - p)^m = (1 - p)^k = P(X > k)$. For an operator: a run of good items does not make a defect any more or less imminent.

Before next week

Attempt exercises 1 to 8. Week 7 asks what these distributions are worth on average: expectation, its properties, and the variance, including proofs of the means and variances quoted in the table above.